

Automated Forgery Detection in Scanned Digital ID Cards Using Boundary Artefact Signatures

Abstract. This paper presents a multi-stage computer vision framework for automated forgery detection in digital scan-copy of an ID-card using boundary artefact signatures generated by copy&Amp;paste tampering. The proposed approach integrates Error Level Analysis (ELA) with a deep learning&Amp;based object detection module to reliably identify and localize manipulated fields in scanned ID-card images. In the first stage, ELA is applied to the input image to produce an ELA-map that highlights potential Copy-Paste Tampering Boundary Artefact (CPTBA) signatures resulting from inconsistencies in DCT-based compression across tampered and untampered regions. In the second stage, a pre-trained YOLOv8 detector is deployed on the ELA-map to automatically localize these artefact patterns. In the final stage, the YOLOv8 detector output is interpreted by a refinement module for generating bounding boxes around suspected manipulated fields leading to precise localization of copy&Amp;paste forgeries. Experimental evaluations demonstrate that the proposed forgery detection framework achieves an accuracy of 99.5% in tampered field detection while significantly outperforming the standalone ELA strategy. The promising outcomes of the proposed work also indicates the effectiveness of boundary artefact signatures as a reliable forensic cue and underscore the potential of the framework for automated digital document integrity verification.

Keywords: Document Forgery Detection · Computer Vision · Machine Learning, · Error Level Analysis (ELA)

1 Introduction

In this current era of information technology, where digital governance is rapidly expanding across various sectors, the verification of scanned copies of important

documents submitted through online portals has become a crucial area of research for developing sustainable engineering solutions. This is particularly important in situations where physical interactions are not feasible, such as during recently passed pandemic scenarios. In many official transactions and banking procedures, scanned copies of intrinsic documents such as ID cards, certificates, bank cheques etc. are used as security documents to verify the authenticity of the users. Consequently, such digital governance systems now faces a grave failure due to submission of counterfeit documents, particularly ID cards as these documents are being digitally manipulated rampantly with malicious intentions. Counterfeiters constantly develop new strategies to exploit weaknesses in such digital governance system while tampering the scanned copies of vital identity documents which are used for authentication purpose. Given the rapid growth in the number of digital copies of ID cards across various levels of official requirements, it is no longer practical to manually verify them within the specified time constraint. Given the growing demand for a reliable and automated authentication tool, it becomes inevitable to conduct research for developing an effective automated verification tool that is capable of detecting tampered scanned copies of intrinsic identity documents.

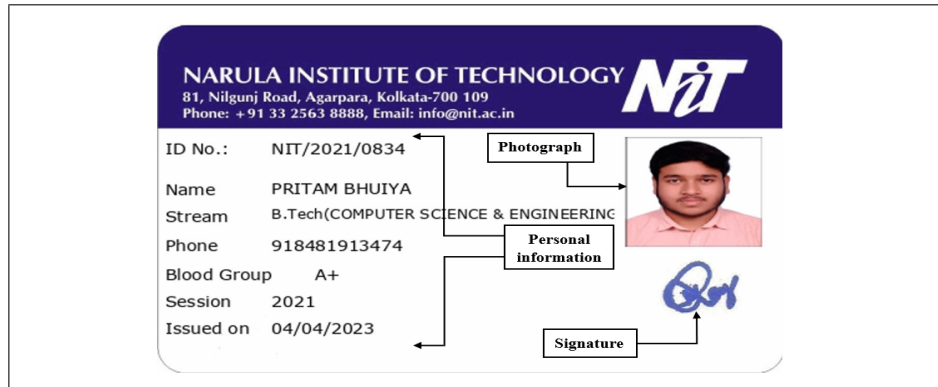


Fig. 1: A valid image of an ID card

In general, counterfeiters employ the copy-paste forgery technique as their main strategy to fabricate ID cards. Such a manipulation strategy involves replacing vital data of an ID card with fake data using digital image editing software such as Adobe Photoshop, GNU GIMP, and other similar programs. With the usage of sophisticated image editing techniques, it becomes exceedingly difficult to visually detect any alterations made to an ID card. In order to obviate these problems, there are conventional methods for document examination that rely on costly specialised equipment and skilled personnel, resulting in a relatively slow and labour intensive process. Some semi-automatic models and machine learning algorithms claim to have the ability to detect forgery, but they still lack the desired effectiveness in terms of making fully automated decisions

with smart intelligence. In recent times, lots of researchers have become highly interested in creating completely automated document verification systems to address the pressing need for managing the increasing number of counterfeit digital copies of ID cards in various government and banking sectors. This urgency also arises from the fact that forgery has become increasingly simple due to rapid technological progress. However, still there is enough scope of research in respect of developing an automated tools for ID card verification in comprehensive manner. In view of this scope of further exploration with the advanced artificial intelligence techniques, we have attempted to develop a deep learning technique based tool for localization of probable tampered regions in scanned digital images of ID cards.

Typically, an ID card contains several information fields prone to tampering, such as Textual Personal Details, the Photograph, and the Signature of the cardholder, as illustrated in Fig. 1. Counterfeiters often target these fields using the widely employed copy-paste forgery strategy. This strategy involves replacing crucial parts of the original image by pasting similar segments from other images. As a result of such forgery strategy, some background inconsistencies emerge around tampered regions of a counterfeit ID card image which we have termed as Copy Paste Tampering Boundary Artefacts (CPTBA) signatures. Our proposed model aims to authenticate an ID card by detecting these CPTBA signatures as shown in Fig. 2.

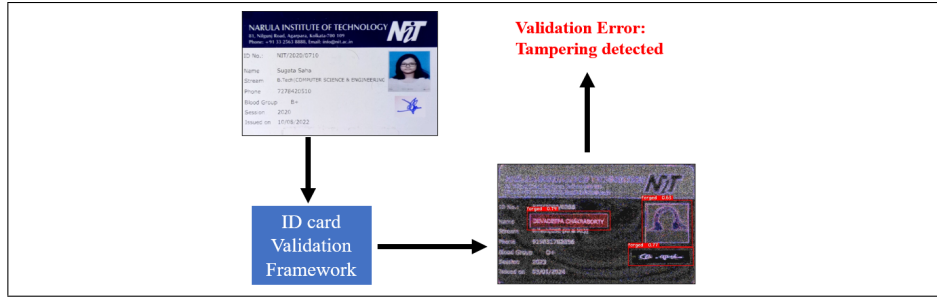


Fig. 2: Automated Detection of a tampered ID card Image

1.1 Objective of the Proposed Framework:

This article intends to present a three-stage computer vision framework based on YOLOv8 (a deep learning tool for object detection) and an ELA module (Error Level Analyzer) that attempts to validate an ID card with respect to the above-mentioned CPTBA signatures that are induced due to cope-paste forgery of crucial ID card fields. During deployment phase, the initial stage of the proposed framework employs Error Level Analysis (ELA) module which looks for different DCT compression levels in the ID card image. In general, a fake ID card image produced by copy-paste forgery gives rise to different DCT compression levels due to the presence of tampered regions which are copied

from different image source. As an output, the ELA module generates an image (ELA image) wherein the CPTBA signatures are highlighted along background inconsistencies between untampered and tampered regions of a counterfeit ID card image. The second stage of the framework employs YOLOv8 based deep learning tool to detect the presence of CPTBA signatures in an ELA image of its respective ID card. The final stage of the framework uses CPTBA detection results produced by YOLOv8 and forms a refined bounding box encasing the tampering field in the ID card image for localization purpose.

During experimentation, we used our own dataset [1] to train and test the framework due to the non-availability of any standard digital ID-card image dataset. The viability of the proposed framework has been established through extensive experimentation with our own dataset while comparing its performance with standalone Error Level Analysis (ELA), a popular copy-paste forgery detection technique.

2 Literature Review

This section presents literature review of the various techniques attempted by researchers for detecting image forgery. The literature review outlines numerous noteworthy attempts by researchers to process copy-paste forgery detection and highlights the scope of research in view of the identified research gap.

2.1 Related Work

Mehrjardi et al. in [11] presented a survey on various forgery detection based on deep-learning methods. However, classical machine learning techniques like SVM were also explored earlier as proposed in the work of Liu, Guangjie, et al.[8] to identify image splicing with artificial blurred boundary based on image edge analysis and blur detection.

Singh et al.[14] makes use of error level analysis (ELA) that points out the difference between the compression rates of authentic and manipulated images. Yang et al.[17] proposes a document forgery detection algorithm based on a deep learning tool (CNN) in fusion with error level analysis (ELA) information. Madake et al.[9] used a combination of Error Level Analysis (ELA) and Metadata Analysis approaches to authenticate the originality of an image passively. They reported an accuracy of 87.24% for the ELA based approach and a accuracy of 73% for the metadata analysis based approach.

The proposed solution by Yap et al. [18] achieves an accuracy of 97.71% and an F1-score of 0.9773 while detecting fake MyKad (Malaysia National Identity card) based on unique colour count, surface roughness score, and local binary patterns. Samarasinghe et al.[13] applied several image-processing techniques effectively for analyzing various security features embedded Sri Lankan Driving License (SLDL) in order to detect its authenticity.

Gonzalez et al.[2] proposed a BasicNet architecture that uses Discrete Fourier Transform to detect image forgery and achieves detection accuracy of 97.5% for real ID card images and a mean of 96.8% for fake ID card images.

2.2 Research Gap and Contributory Scope of the Proposed Work

Despite the extensive research on general image forgery detection, our literature review indicates a notable gap in exploring deep learning techniques specifically for addressing the problem of forgery detection in digital identity (ID) cards. These documents typically contain several critical information fields—such as textual personal details, the photograph, and the cardholder’s signature—that are particularly vulnerable to tampering. A common manipulation technique employed by counterfeiters is copy-paste forgery, in which these critical information fields of an original ID card image are replaced with visually similar segments sourced from other images. This method often introduces background inconsistencies around the tampered regions, which we termed as Copy-Paste Tampering Boundary Artefacts (CPTBA).

To address this challenge, we propose a novel three-stage computer vision framework that leverages the YOLOv8 deep learning model for object detection, in conjunction with an Error Level Analysis (ELA) module. Unlike conventional forgery detection approaches, our method utilizes ELA-transformed images of counterfeit ID cards, annotated with CPTBA signatures surrounding the manipulated fields, to train the YOLOv8 model. During the deployment phase, the trained YOLOv8 model attempts to identify the tampered regions by detecting the presence of CPTBA signatures.

3 Proposed Methodology

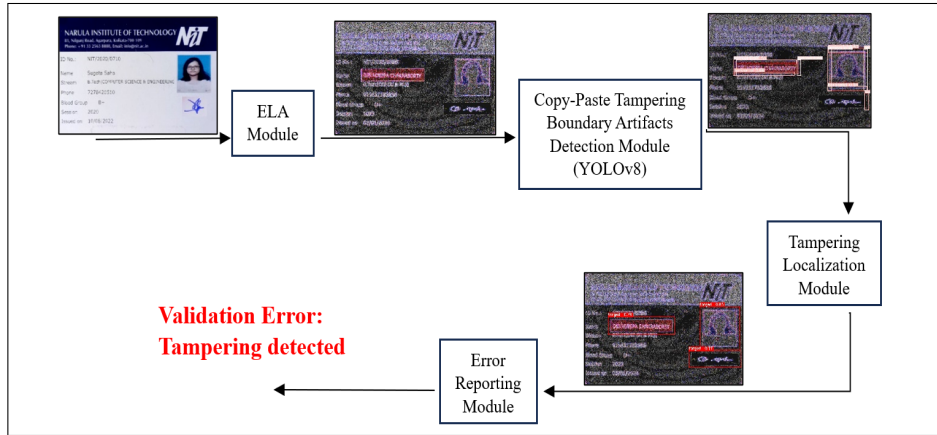


Fig. 3: Overview of the process flow of the proposed framework

This section offers a comprehensive overview of the methodology of our proposed work that aims to recognise forged ID cards by detecting Copy Paste Tampering Boundary Artefacts (CPTBA). Fig. 3 outlines the process-flow of

the proposed framework. It accepts an ID card image as input and passes it through a three-stage framework consisting of various modules with each having distinct role. This framework identifies a fake ID card manipulated with copy-paste forgery and marks its tampered fields by generating bounding boxes around them. As previously mentioned, the proposed framework operates primarily in three stages – ELA-Image Generation Stage, YOLOv8 CPTBA Detection Stage, and Tampering Field Localization Stage. The details of each stage, along with some additional background preliminaries related to ELA and YOLOv8, are presented below to illustrate the methodology of the proposed approach.

3.1 Stage-1: ELA-Image Generation Stage

The proposed framework incorporates the ELA module as a pre-processing technique during the initial stage that utilizes Error Level Analysis (ELA) strategy for generating ELA-image with an aim to highlight CPTBA signatures in a fake ID card image. Error Level Analysis (ELA) is a forensic technique used in digital image analysis for detecting alterations or manipulations, often employed in forgery detection. This method relies on the principle that when an image is saved or compressed multiple times, the error levels introduced during each compression step are not uniform across the image. In other words, areas that have been modified or tampered with copy-paste operations tend to exhibit different error levels compared to non-tampered areas.

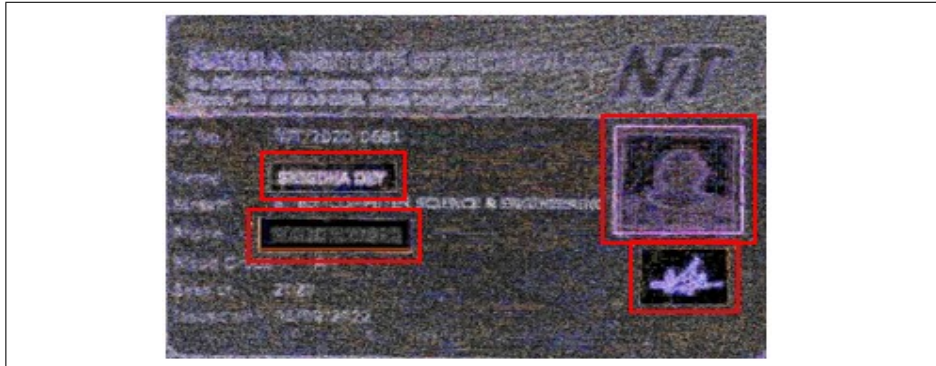


Fig. 4: ELA image of a Tampered ID card

In practice, ELA involves compressing an image multiple times with DCT based compression techniques like JPEG baseline strategy and then analyzing the error levels introduced during compression. Typically, at initial stage, the candidate image is saved at a relatively high quality compression setting to minimize compression artifacts. Then, the same image is re-saved at a lower quality compression setting, which introduces more compression artifacts. The difference in error levels between these two versions of the image is calculated

and visualized, often as a gray-scale image termed as ELA-image. The resulting ELA-image displays variations in error levels across different areas of the image and the areas with high variations indicate the presence of CPTBA signatures around potential tampered regions as highlighted in Fig. 4. The overall process of ELA-image generation is summarized below.

Algorithm 1 Procedure for Generation of ELA-image

Input: Original image I , Compression quality Q

Output: Highlighted ELA-image E_{otsu}

- 1: {Step 1: Accept Input and initialization}
 - 2: Load original image I
 - 3: {Step 2: Re-save Image}
 - 4: Re-encode I using compression quality Q to create $I_{resaved}$
 - 5: Save $I_{resaved}$ as a temporary buffer/file
 - 6: {Step 3: Create ELA-image}
 - 7: **for** each pixel (x, y) **do**
 - 8: $D(x, y) \leftarrow |I(x, y) - I_{resaved}(x, y)|$ {Pixel-wise subtraction}
 - 9: **end for**
 - 10: Convert absolute difference D to grayscale range $[0, 255]$ to obtain E_{gray}
 - 11: $E_{otsu} \leftarrow \text{ApplyOTSUThreshold}(E_{gray})$ {Highlight significant differences}
 - 12: {Step 4: Produce Output}
 - 13: **return** E_{otsu}
-

3.2 Stage-2: YOLOv8 CPTBA Detection Stage

The proposed framework employs the You Only Look Once (YOLOv8) based object detection tool, with CSPDarknet-53 serving as the underlying architecture. YOLOv8, a prevalent deep neural network model developed, is used in our proposed framework to automatically detect the highlighted areas in an ELA-image which are indicative of CPTBA signatures.

In our proposed framework, YOLOv8 model uses CSPDarknet-53 architecture (Fig. 5) as its backbone which is a series of convolutional layers followed by downsampling and residual connections []. The primary function of the backbone network is to extract hierarchical features from the input image. These features represent different levels of abstraction, ranging from low-level features like edges and textures to high-level features like object shapes and patterns. CSPDarknet-53 utilizes a series of convolutional and pooling layers to progressively downsample the spatial dimensions of the feature maps while increasing the number of channels, effectively capturing features at multiple scales.

The output of the backbone network is passed to a detection head, which is responsible for predicting bounding boxes and class probabilities for objects present in the input image. YOLOv8 employs a single detection head, which

consists of convolutional layers followed by a final prediction layer. The prediction layer predicts bounding boxes, confidence scores, and class probabilities for a predefined set of anchor boxes at each spatial location in the feature map.

YOLOv8 divides the input image into a grid of cells. Each cell is responsible for predicting bounding boxes for objects whose center falls within that cell. Anchor boxes are predefined bounding box shapes of different sizes and aspect ratios. YOLOv8 predicts bounding box coordinates and dimensions relative to these anchor boxes. Each grid cell predicts multiple bounding boxes (usually 3-5) along with their confidence scores and class probabilities.

After prediction, YOLOv8 employs non-maximum suppression (NMS) to remove redundant bounding box predictions. NMS filters out overlapping bounding boxes based on their confidence scores and IoU (Intersection over Union) threshold, keeping only the most confident predictions for each object. The final output of YOLOv8 CSPDarknet-53 architecture is a set of bounding boxes, each associated with a class label and confidence score, representing the detected objects in the input image.

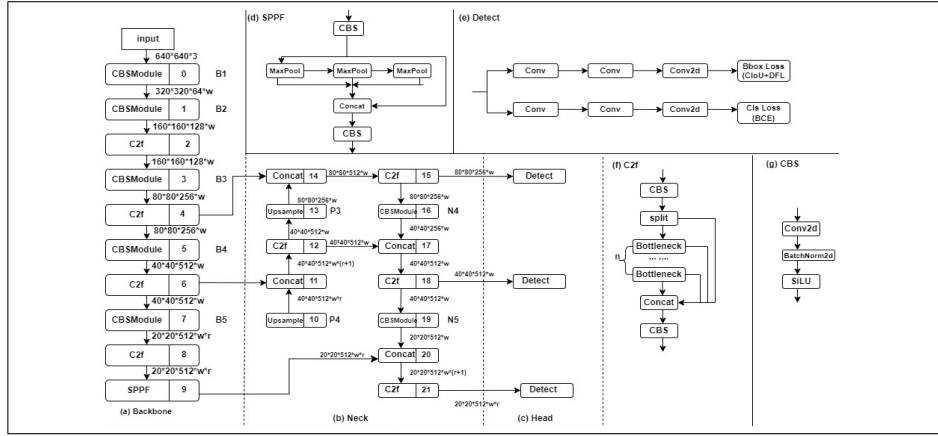


Fig. 5: Overview of the YOLOv8 architecture

Development of YOLOv8 Model for detection of CPTBA Signatures

in ELA-Image: In our proposed framework, the YOLOv8 model is developed using ELA-images of various fake ID cards whose crucial fields are manipulated by copy-paste forgery techniques. The highlighted areas in an ELA-image which are indicative of CPTBA signatures are annotated by drawing rectangular box as shown in Fig. 7. The YOLOv8 model is trained with these annotated ELA-images so that it can locate the CPTBA signatures around the tampered fields of a fake ID card image. As stated earlier, YOLOv8 model takes the ELA-image of an ID card as input and attempts to enclose the highlighted CPTBA signatures around the tampered fields with four rectangular boxes. It is often found that the

counterfeiter makes rectangular copy-paste forgery wherein a crucial field in the target ID card image is replaced by a similar rectangular image field sourced from another image. This rectangular copy-paste forgery tends to produce CPTBA (Copy Paste Tampering Boundary Artefacts) signatures around a tampered field. In fact, the inconsistency between the backgrounds of source and target image causes such boundary artefacts around the four sides of the pasted field in the target ID card image. Our YOLOv8 model attempts to detect these four sides of a tampered field in an ELA-image which are likely to be highlighted by ELA technique as explained previously.

3.3 Stage-3: Tampering Field Localization Stage and Authentication Error Reporting

In final stage, the detection results of the YOLOv8 model are analyzed to create bounding boxes around tampered fields of a ID card and the error reporting module produces authentication error in case the framework identifies any tampered field in the ID card image. As illustrated earlier that our YOLOv8 model attempts to detect the CPTBA signatures around a tampered field by encapsulating them within rectangular boxes. The localization module further attempts to locate four bounding boxes (top, bottom, left and right) around a crucial field produced by YOLOv8 model. If the localization module identifies at least two bounding boxes surrounding any crucial field, it encloses the entire field within a rectangular box and notifies the authentication error generation module of potential copy-paste forgery with respect to the corresponding ID card field. Effectively, the rectangular bounding boxes generated by the localization module help us visualize the tampered fields within the fake ID card. The possible cases to be dealt with the localization module along with its outcomes are depicted in Fig. 6.

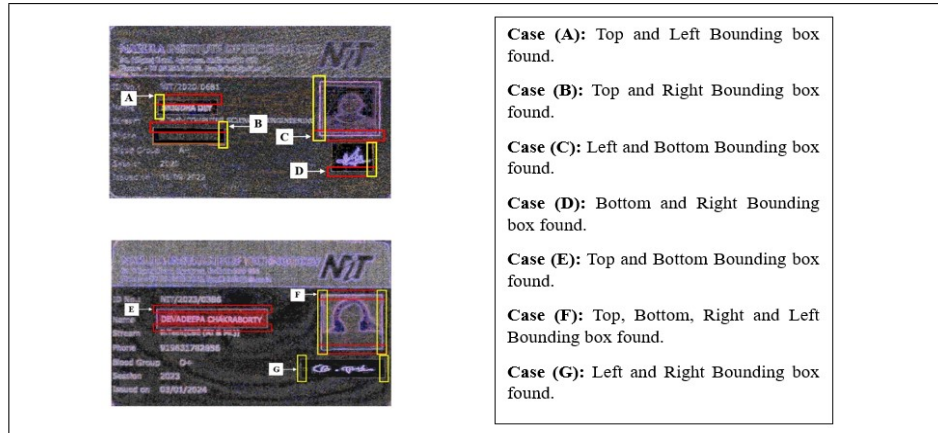


Fig. 6: Cases for localizing Bounding boxes

4 Experimental Results and Discussion

This section presents our experimental configuration details and reports about the findings of our experimentation with proposed framework along with performance analysis. As experimental environment, we have worked with Google Colaboratory Platform having Tesla T4 GPU with 32 GB of RAM. The detailed description of the dataset used in our experiment is provided below.

4.1 Dataset Description

In order to analyze the performance of our proposed framework, we have prepared a unique dataset consisting of 200 scanned images of collage ID cards with identical physical dimensions. The experimental dataset along with the corresponding annotation metadata is made available at [1] for substantiation. In order to keep variations in the dataset, the ID card images were acquired using different cameras under varying illumination conditions. Subsequently, each ID card image undergoes scaling operation to derive three more images with resolution 900×500 , 675×375 , and 450×250 . As a result, three categories of dataset with each category having specific image resolution are generated for our experimentation. In each category of dataset, the crucial fields of an ID card image (target) such as Textual Personal Details, the Photograph, and the Signature of the cardholder are randomly replaced by similar fields by copying them from some other ID card images. This random copy-paste forgery results in generation of tampered ID photographs exhibiting diverse forms of forgeries. Fig. 7 displays some tampered ID card images with annotations in tampered fields.

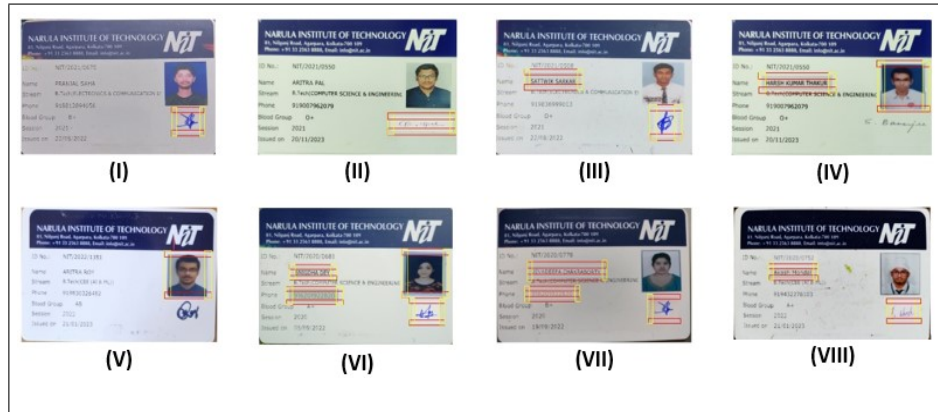


Fig. 7: Sample images of Different Tampered ID cards

In our experimentation, each dataset category consists of one hundred authentic photos and one hundred tampered photos of identity cards. Among the tampered photo collection, twenty-five percent of ID cards contain only fabricated textual personal information, twenty-five percent contain solely forged

photos of the cardholder, twenty-five percent feature only counterfeit signatures, and the remaining twenty-five percent exhibit a combination of these three types of forgeries. In order to develop our YOLOv8 model, all the manipulated ID card images are annotated with rectangular bounding boxes, as depicted in Figure 4 and 85% of the dataset for training purposes, while the remaining 15% was reserved for testing. Fig. 8 illustrates the training loss graph and distribution summary of the dataset.

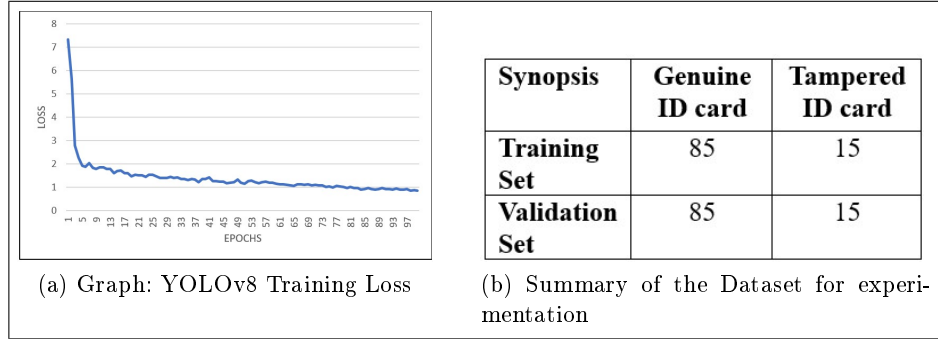


Fig. 8: Loss Graph During Training Phase of YOLOv8 Model and Synopsis of the dataset

4.2 Performance Evaluation of the Proposed Model

The performance of the proposed model is assessed in terms of IOU (Intersection over Union) metric and forgery detection accuracy. The IOU metric is evaluated based on the proportion of pixels that are covered by both the manually annotated tampered area and the predicted tampered area. The IOU can be computed using Eqn. 1 where $Predict_{BB_{ox}}$ represents the rectangular bounding box that encloses the tempered region predicted by proposed model and $GT_{BB_{ox}}$ denotes the ground-truth based rectangular bounding box enclosing the actual tampered region in the ID card image. It is worth noting that a higher IOU value indicates enhanced localization accuracy of the proposed framework.

$$IOU = \frac{Predict_{BB_{ox}} \cap GT_{BB_{ox}}}{Predict_{BB_{ox}} \cup GT_{BB_{ox}}} \quad (1)$$

Table 1 shows a few results of YOLOv8 and Localization module of the proposed framework for its qualitative assessment. The performance of our forgery field localization module can also be assessed from the listed IOU values. The first column of Table 1 displays a few samples of ID card photos with ground truth annotations of tampered fields. The second column shows the respective ELA-images with CPTBA-detection results of our YOLOv8 model. In subsequent columns, the outcomes of the localization module along with respective IOU values are provided for visual inspection.

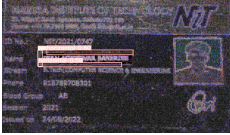
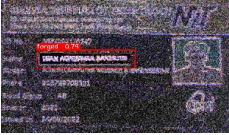
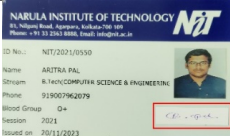
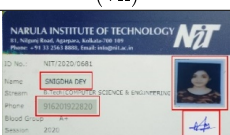
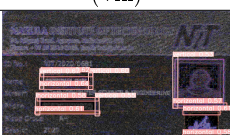
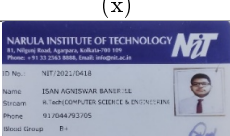
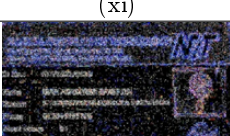
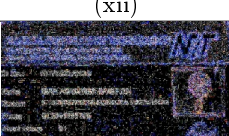
Sl. NO.	Ground Truth	YOLOv8 Module	Localization Module	IOU (%)
1.	 (i)	 (ii)	 (iii)	92
2.	 (iv)	 (v)	 (vi)	87
3.	 (vii)	 (viii)	 (ix)	95
4.	 (x)	 (xi)	 (xii)	85
5.	 (xiii)	 (xiv)	 (xv)	100

Table 1. Results of YOLOv8 and Tampering Field Localization Module of the Proposed Framework

The detection accuracy of tampered regions in Identity Card images is computed depending on True Positive (TP), True Negative (TN), False Positive (FP), and False Negative (FN) cases, which are determined according to the following criteria.

- True Positive (TP): If our model can successfully locate all tampered fields in an identity card, it will be treated as a True Positive case.
- True Negative (TN): If our model correctly identifies a genuine identity card in absence of any tampered field, it will be treated as True Negative case.

- False Positive (FP): If our model incorrectly locates any tampered field in a genuine identity card, it will be treated as False Positive case.
- False Negative (FN): False Negative cases refer to the situations, when our model fails to locate any tampered field in a forged identity card.

The forgery detection accuracy and precision values of the proposed framework have been measured using Eqn. 2 and Eqn. 3 respectively. It may be noted that an increase in the True Positives (TP) and True Negatives (TN) leads to an improvement in the overall detection accuracy of the model.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (2)$$

$$Precision = \frac{TP}{TP + FP} \quad (3)$$

Table 2 presents quantitative evaluation of the proposed framework with respect to its forgery detection ability based on True positive, True negative, False positive, and False negative cases as shown in Table 3. In order to make a comparative study, the forgery detection results are also obtained for standalone ELA strategy while varying image resolutions of the ID cards. Our proposed framework demonstrates better performance in terms of precision and accuracy across all the three categories of datasets with varying dimensions. The experimental results demonstrate that the proposed framework achieves better detection accuracy and precision values while working with higher image resolutions (99.5% for 900×500) and it outperforms the standalone ELA technique which is widely used for general image forgery detection.

Table 2. Quantitative Assessment: Forgery Detection Results

Various Methods	Image Scale	Average Precision	Average Accuracy
ELA [14]	450×250	65.53%	71.9%
Proposed Method	450×250	81.6%	83.32%
ELA [14]	675×375	69.45%	77.23%
Proposed Method	675×375	87.3%	94.76%
ELA [14]	900×500	85.76%	91.13%
Proposed Method	900×500	100%	99.5%

Table 3. Confusion Matrix for ID Card Images (900×500)

	Predicted Tampered	Predicted Genuine
Actual Tampered	TP: 147	FN: 1
Actual Genuine	FP: 0	TN: 52

5 Conclusion

This article aims to present a multi-stage computer vision framework that employs deep learning tool in fusion with Error Level Analysis (ELA) technique

to verify the authenticity of a scanned image of ID card. The proposed framework attempts to detect Copy Paste Tampering Boundary Artefacts (CPTBA) signatures that are generated due to tampering of crucial fields in the ID card. During the deployment phase, the proposed framework applies an Error Level Analysis (ELA) module to scanned image of an identity card that produces a special image as output termed as ELA-image. This ELA-image highlights the differences between authentic and manipulated images by analyzing the error levels introduced during DCT based compression to the input image. In the second phase of the framework, YOLOv8, a deep learning based object detection tool, is employed to ELA-image for detecting the existence of CPTBA signatures on an ID card. In order to localize the manipulated fields of the ID card, the final phase of the framework forms a bounding box surrounding the tampered area in the ID card image by analyzing the detection results of YOLOv8. The experimental results are found to be promising which indicate the effectiveness of the proposed framework in terms of localizing the tampered fields in an ID card with respect to CPTBA signatures.

While the proposed framework yields encouraging results in the detection and localization of tampered regions within ID card images, its development and evaluation revealed several limitations and challenges. These encompass data-related constraints, technical shortcomings, and issues concerning model's generalization ability.

The current scope of our proposed framework is limited to rectangular copy-paste forgery of digital identity cards. In future, the effort may be directed towards resolving the authentication issues of document-images which are tampered with copy-paste forgery of non-rectangular regions. Additionally, the core working principle of the proposed framework may further be explored to probe its forgery detection capabilities while dealing with deep-fake photos as well as videos.

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